LECTURE 7: SCIENCE AND TECHNOLOGY IN MALAYSIA

4/5/2025

MOSTI Funds & Grant

Dana pembiayaan projek untuk meningkatkan pertumbuhan ekonomi negara dan memberi kesejahteraan kepada masyarakat.

More Information







MOSTI



MOSTI stands for the **Ministry of Science, Technology and Innovation** in Malaysia.

It is a key government ministry responsible for driving the country's agenda in:

- Science and research development
- Technology advancement and innovation
- Digital economy and emerging technologies
- National policies related to R&D, including areas like AI, biotechnology, and clean energy

MOSTI



Some well-known agencies under MOSTI include:

Malaysian Industry-Government Group for High Technology (MIGHT)

Malaysian Space Agency

National Institutes of Biotechnology Malaysia

1. MALAYSIAN INDUSTRY-GOVERNMENT GROUP FOR HIGH TECHNOLOGY (MIGHT)

The **Malaysian Industry-Government Group for High Technology (MIGHT)** is a strategic agency in Malaysia that connects **government, industry, and academia** to drive high-technology development.

MiGHT
Malaysian Industry-Government Group for High Technology

WHAT MIGHT DOES

MIGHT acts as a **think tank and facilitator,** focusing on:

Strategic foresight and policy advice for emerging technologies

Technology development and commercialization

Industry collaboration, especially in high-impact sectors

Public–private partnerships to accelerate innovation

KEY FOCUS AREAS

MIGHT is involved in sectors such as:

Advanced manufacturing

Aerospace and space technology

Green technology and sustainability

Digital and emerging technologies (AI, IoT, etc.)

ROLE IN MALAYSIA

MIGHT operates under the **Ministry of Science, Technology and Innovation (MOSTI)** and plays an important role in:

Supporting Malaysia's **innovation ecosystem**

Helping industries adopt **future technologies**

Strengthening Malaysia's competitiveness in the global high-tech landscape

2. MALAYSIAN SPACE AGENCY

The **Malaysian Space Agency (MYSA)** is Malaysia's national agency responsible for space-related activities and development.

What MYSA does

- MYSA focuses on using **space science and technology** for national development, including:
- **Satellite development and operations**
- **Earth observation** (monitoring environment, forests, oceans, disasters)
- **Space research and innovation**
- **Geospatial data and remote sensing services**

MALAYSIAN SPACE AGENCY

Why it matters

1. MYSA plays a practical role in supporting Malaysia's needs, such as:
2. Disaster management (flood monitoring, haze tracking)
3. Agriculture and food security
4. Urban planning and environmental protection
5. National security and surveillance

3. NATIONAL INSTITUTES OF BIOTECHNOLOGY MALAYSIA (NIBM)

The **National Institutes of Biotechnology Malaysia (NIBM)** is a national agency that coordinates and promotes **biotechnology development** in Malaysia.

NIBM serves as a **platform for collaboration and research coordination** in biotechnology. Its main roles include:

1. **Strengthening biotech research** across universities and institutes
2. **Facilitating collaboration** between academia, industry, and government
3. **Supporting commercialization** of biotech innovations
4. **Developing human capital** in biotechnology fields

NIBM
NATIONAL INSTITUTES OF BIOTECHNOLOGY MALAYSIA

NATIONAL INSTITUTES OF BIOTECHNOLOGY MALAYSIA (NIBM)

Key focus areas

1. NIBM focuses on several important sectors:
2. Healthcare and medical biotechnology (e.g., vaccines, diagnostics)
3. Agricultural biotechnology (e.g., crop improvement, food security)
4. Industrial biotechnology (e.g., bio-based products, enzymes)
5. Environmental biotechnology (e.g., waste treatment, sustainability)

NIBM
NATIONAL INSTITUTES OF BIOTECHNOLOGY MALAYSIA

MAJOR UNIVERSITIES OFFERING S&T PROGRAMMES

MAJOR UNIVERSITIES OFFERING S&T PROGRAMMES

Malaysia has a strong network of public and private universities with S&T specialisations:

Public Universities

Universiti Malaya (UM)

- AI, data science, engineering, biotechnology, science & technology studies

Universiti Kebangsaan Malaysia (UKM)

- Environmental science, nuclear science, technology policy

Universiti Putra Malaysia (UPM)

- Agricultural technology, biotechnology, food science

Universiti Teknologi Malaysia (UTM)

- Engineering, smart technology, robotics

Universiti Sains Malaysia (USM)

- Pure sciences, sustainability, innovation

Private Universities

Multimedia University (MMU) – ICT, digital technology

Taylor's University – applied sciences, tech innovation

Sunway University – data science, environmental tech

KEY STI PILLARS AND FRAMEWORKS

Malaysia Perspective

KEY STI PILLARS AND FRAMEWORKS

National Science, Technology and Innovation Policy 2021–2030 (NSTIP): Anchors Malaysia's development, focusing on strengthening technological self-reliance and reducing foreign dependency.

MySTIE 10-10 Framework: Links ten global science drivers with ten socioeconomic drivers to foster economic impact.

Sector Roadmaps: Dedicated strategies for Artificial Intelligence (2021–2025), Electrical & Electronics, Advanced Materials, and Blockchain.

MAJOR TECH INDUSTRIES

Semiconductors & Electronics: Malaysia is a global hub for semiconductor devices and electrical goods manufacturing.

Digital Economy: Rapid adoption of 5G, robotics, and digital payments is accelerating industrial digital transformation.

Biotechnology & Agriculture: Emphasis on biotechnology and enhancing agricultural productivity through innovation.

RESEARCH, INNOVATION, AND TALENT

Performance: High-tech exports account for over 50% of manufactured exports.

Research & Development (R&D): Major efforts focus on bridging the gap between research and industrial application, with MASTIC acting as the central STI information provider.

Challenges: There is a recognized need to increase the number of students graduating in STEM fields (engineering, IT) to support industry demands.

KEY INSTITUTIONS AND INITIATIVES

Ministry of Science, Technology and Innovation (MOSTI): Sets national policy, focusing on ICT, biotech, and space technology.

ASTI (Association of Science, Technology & Innovation): Promotes science education through initiatives like the Young Inventors Challenge.

MUST (Malaysia University of Science and Technology)**:** Provides specialized education to develop high-level technical talent

MAJOR ROADMAPS AND INITIATIVES:

AI Roadmap 2021–2025: Developing artificial intelligence capabilities.

Electrical and Electronics Roadmap 2021–2030: Advancing the semiconductor sector.

National Robotics Roadmap 2021–2030: Driving automation and robotics adoption.

National Advanced Materials Technology Roadmap 2021–2030: Enhancing industrial materials.